

Lecture 9: Limiting CO₂ emissions

DTU Course 46770: Integrated Energy Grids

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Problem 9.1. Producing electricity in coal and gas power plants entails the efficiency, CO₂ emissions and fuel cost shown in Table 1. What is the cheapest option to produce electricity? Which CO₂ tax would be necessary to alter the merit order of the two technologies and incentivise lower CO₂ emissions per MWh of produced electricity?

	Coal Power plant	Combined Cycle Gas Turbine (CCGT)
Fuel cost (USD/MWh thermal energy)	6.2	20.1
Efficiency of power plant (MWh electricity / MWh thermal energy)	0.33	0.59
Emissions (tCO ₂ /MWh thermal energy)	0.336	0.198

Table 1: Electricity production in coal and gas power plants.

Problem 9.2. Build the model of Portugal described in Problem 3.1 in PyPSA and assume that methane gas emits 0.198 tCO₂ per MWh of thermal energy contained in the gas.

Consider the annualised capital costs and marginal generation costs for the different technologies in the following table.

Technology	Annualised capital costs (EUR/MW/a)	Marginal generation costs (EUR/MWh)
Onshore Wind	101,644	0
Solar PV	51,346	0
OCGT	47,718	64.7
CCGT	104,788	46.8
Battery inverter	24,678	0
Battery energy capacity	12,894	0

Table 2: Costs assumptions.

Limit the maximum CO₂ emissions to 5 MtCO₂/year.

- Calculate the optimal installed capacities and plot the hourly generation and demand during January.
- What is the CO₂ tax required to meet this CO₂ emission limit?

Problem 9.3. Build the model described in Problem 3.2 in PyPSA (i.e. the problem above but not including the CCGT generator with exogenous capacity).

- Solve the problem for different CO₂ values ranging from 5 MtCO₂/year to zero. Plot the total system cost and the required CO₂ prices as a function of the emissions allowance.
- Show what would happen if a CO₂ tax was used instead of a CO₂ limit. You can do this by adding a carbon tax to the relevant generators, and then comparing your results (total system costs, emissions, and generation mix) with the previous section. Make sure to use your results from part (a) for the appropriate carbon taxes.